LESSON 3.3

Determine Equivalent Ratios Using Tables

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use a ratio table to find equivalent ratios.

WHY THIS MATTERS Brian will use the recipe shown to make clay. He needs to make multiple batches of clay. How can Brian determine how much baking soda and cornstarch he needs for multiple batches of clay to get the same consistency?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Ratio Tables *Spanish: Tablas de razones*	Tables that organize pairs of equivalent ratios.	2:3, 4:6, and 6:9 belong in the same ratio table.
Ratio table *Spanish: Tabla de razones*	A table of equal ratios that shows a pattern.	5, 10, 15 / 2, 4, 6
Equivalent ratio *Spanish: Razón equivalente*	Two ratios that mean the same thing, like 1:2 and 2:4.	2:3 = 4:6 = 6:9
Scale factor *Spanish: Factor de escala*	The number you multiply both parts of a ratio by to get an equal ratio.	2:3 × 2 → 4:6 (scale factor is 2)
Pattern *Spanish: Patrón*	A rule that numbers follow again and again.	Each row adds 2 cups juice and 3 cups water
Additive pattern *Spanish: Patrón aditivo*	Building new rows of a ratio table by adding each column's own constant amount, so the two columns grow together.	2:3 → 4:6 → 6:9 (add 2 and 3 each time)

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A mix uses 5 scoops of powder for every 2 cups of water. How much water for 15 scoops?

1. Find the factor: 15 ÷ 5 = 3.
2. Multiply the water by 3 too: 2 × 3 = 6.
3. Check: 15 : 6 ÷ 3 = 5 : 2.

Answer: 15 scoops go with 6 cups of water

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A drink uses 1 cup of juice for every 4 cups of water, so the ratio is 1:4.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I multiplied both terms by _____ .
- _____ and _____ are equivalent because _____ .
- Adding does not work because _____ .
- The factor is _____ because _____ × _____ = _____ .

Word bank equivalent | ratio table | factor | multiply | divide | both | same | scale up | scale down | for every | row | pattern

WATCH OUT!

A common mistake

A common mistake in Ratio Tables is scaling only ONE part of the ratio, or adding the scale factor to a part instead of multiplying by it

LESSON 3.3

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Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

A cookie recipe uses 2 cups of butter for every 5 cups of flour. Complete the ratio table.

Batches	Butter (cups)	Flour (cups)
1	2	5
2		
3		

HINT Row 1 shows ONE batch: 2 cups butter and 5 cups flour. Every other row is copies of that batch.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A recipe uses 3 eggs for every 5 cups of flour. How many eggs are needed for 15 cups of flour?

(A) 6
(B) 9
(C) 10
(D) 12

HINT Notice the flour jumped from 5 cups to 15 cups — the eggs must grow by the same factor.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which row does NOT belong in a ratio table for 4:5?

(A) 6:10
(B) 8:10
(C) 12:15
(D) 16:20

HINT Every row in a 4:5 table must simplify back to exactly 4:5.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A ratio table shows 3:7, 6:14, and 9:21. What is the next row?

(A) 10:24
(B) 12:28
(C) 12:24
(D) 15:28

HINT Look down each column: how does 3 → 6 → 9 change? How does 7 → 14 → 21 change?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A ratio table for 2:5 starts with 2:5, 4:10. What is the third row?

(A) 5:10
(B) 6:15
(C) 6:12
(D) 8:15

HINT Compare row 1 (2:5) with row 2 (4:10). What happened to each column?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Catch Priya's Mistake

1. **Read the order** A trail mix recipe uses 2 cups of nuts for every 5 cups of raisins (2:5). Priya wants to make a bigger batch using a scale factor of 4.
2. **Pick a scale factor** Priya's scale factor is 4. A scale factor should multiply BOTH parts of the ratio.
3. **Scale the nuts** 2 × 4 = 8 cups of nuts
4. **Scale the raisins** 5 + 4 = 9 cups of raisins

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Look at Step 4. Priya turned 5 into 9, but is 8:9 the same ratio as 2:5?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you find each new row in the pancake ratio table? If someone asked for 10 batches instead of 4, would counting up by adding 2 and 3 each row still be a fast way to find the answer?

SENTENCE STARTER I found the 4-batch row by multiplying ___ and ___ by ___, which is faster than adding ___ and ___ over and over ___ times.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.3

Determine Equivalent Ratios Using Tables

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the ratio table for lemon juice to water in Chef Reyes's lemonade recipe (1 cup lemon juice for every 4 cups water).

Pitchers	Lemon Juice (cups)	Water (cups)
1	1	4
2		
5		
10		

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Which rows belong in a ratio table for 3:4? Sort them into correct and incorrect.

Groups: Belongs in 3:4 Table | Does NOT Belong

_____ 6:8

_____ 9:12

_____ 15:20

_____ 6:10

_____ 12:14

_____ 9:16

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

In a ratio table, the ratio of sugar to butter is 3:2. If you use 12 cups of sugar, how many cups of butter do you need?

(A) 6
(B) 8
(C) 9
(D) 10

PER 1

4 **Circle the letter of the best answer. Show how you know.**

A ratio table shows 5:8, 10:16, 15:24. What scale factor was used from the first row to the third row?

(A) 2
(B) 3
(C) 5
(D) 8

PER 1

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Catch Marcus's Mistake

1. **Read the table** A ratio table starts with 3:8. Marcus needs to find the row where the first value is 12.
2. **Find the change** Marcus sees the first value went from 3 to 12, a difference of 9.
3. **Apply the change** Marcus adds 9 to the second value: 8 + 9 = 17
4. **Write the new row** Marcus's answer: 12:17

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you find each new row in the pancake ratio table? If someone asked for 10 batches instead of 4, would counting up by adding 2 and 3 each row still be a fast way to find the answer?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.3

Determine Equivalent Ratios Using Tables

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Chef Reyes needs to scale a soup recipe. The original uses 7 cups of broth for every 4 cups of vegetables. Complete the table, then explain the pattern you see.

Batches	Broth (cups)	Vegetables (cups)
1	7	4
2		
5		
0.5		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find Devon's Mistake

1. **Read the problem** A recipe uses 5 cups of oats for every 2 cups of honey. Complete the table for 3 batches.
2. **Find the scale factor** 3 batches → multiply by 3
3. **Scale the oats** 5 × 3 = 15 cups of oats
4. **Scale the honey** 2 + 3 = 5 cups of honey

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A recipe uses 4 cups of flour for every 3 cups of sugar. A student filled in a ratio table: 4:3, 8:6, 12:9, 14:12. Find and fix the error in the last row. Explain what went wrong.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How did you find each new row in the pancake ratio table? If someone asked for 10 batches instead of 4, would counting up by adding 2 and 3 each row still be a fast way to find the answer?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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